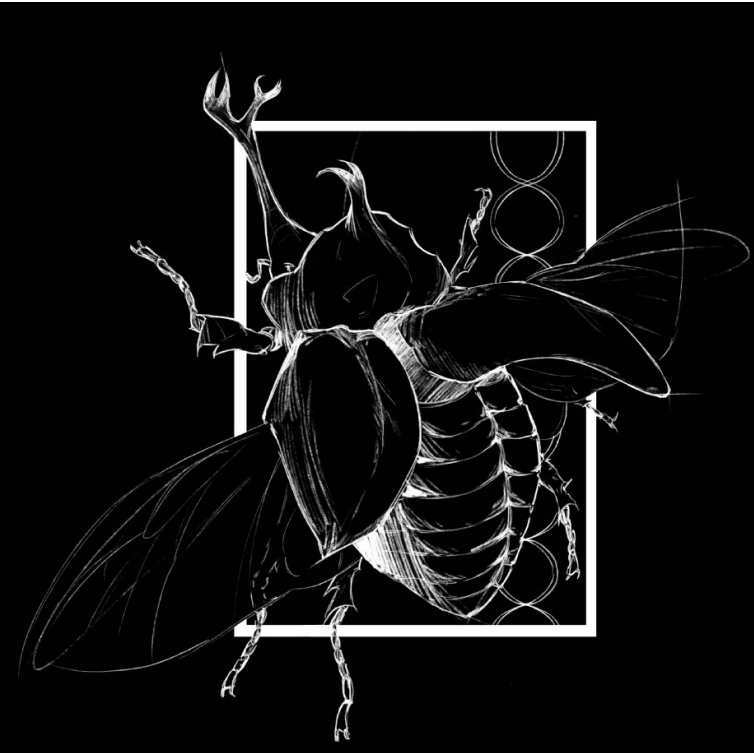




Background

Chromosome evolution is an essential driving force of speciation through fusions, fissions, and polyploidy. Typically, a faster rate of chromosome evolution indicates rapid changes in a species' genome. Lepidoptera have holocentric chromosomes that contain multiple kinetochores along the length of the chromosome, creating more opportunities for fragmentation and rearrangement without losing genetic material, allowing species to evolve faster. For these changes to become permanent characteristics of a species, they must undergo the process of fixation. However, according to the Drift Barrier Hypothesis, whether a population undergoes fixation heavily depends on its size. In smaller populations, such as specialists, fusions and fissions within holocentric chromosomes contribute to higher rates of fixation, while in larger populations the rate is expected to be lower. Specialists typically form small, fragmented populations due to their dependence on patchily distributed host plants, whereas generalists maintain larger, more connected populations with higher gene flow. This difference in population structure creates the conditions necessary for testing whether ecological factors drive differential rates of chromosome evolution. **The goal of this project is to analyze how the rate of chromosome evolution is impacted by diet specialization and generalization in Papilionoidea by testing the hypothesis that specialization accelerates the rate of chromosome evolution through chromosomal rearrangements that facilitate fixation.**

Study Design & AI

- Guidance from Dr. Blackmon and Gemini 2.5 Pro were used to brainstorm and research Lepidoptera. Gemini was used to aid in the research of diet specialization and generalization, and chromosome evolution.
- Newick and CSV files for Lepidoptera haploid number were provided by Dr. Heath Blackmon as part of the Blackmon Lab. The Papilionoidea phylogenetic tree and diet data was obtained from Barnabas H. Daru *Data from: A global database of butterfly species native distributions*. These two data sources were used to create Figure 1 along with code in R from Claude Opus 4.5 and guidance from Dr.Blackmon.
- The density vs rate difference graph (figure 2) and data table (table 1) were created and analyzed using code provided by Dr. Blackmon and bug fixes with Claude Opus 4.5. The analysis uses ChromePlus in R.
- A total of 500 Papilionoidea were used in the analysis. Lepidoptera, specifically Papilionoidea are ideal for this study because they contain holocentric chromosomes, diverse dietary strategies, and an established phylogeny. Specialists were defined by feeding on a single plant genus



Model Fitting

Chromosome evolution was modeled using the ChromePlus package in R, which analyzes chromosome number changes alongside a binary trait. This model separates rates for ascending changes (fissions), descending changes (fusions), and polyploidy events. Rates were estimated for both the specialist and generalist lineages for a direct comparison. A Bayesian Markov Chain Monte Carlo (MCMC) approach was employed using a diversitree package, with 1000 iterations and the first 25% discarded as burn in convergence. The rate difference was calculated by subtracting generalists rates from specialists rates for each type of chromosomal change. Statistical significance was assessed using 95% credible intervals, where any intervals excluding zeros indicate a significant difference in chromosome evolution rate between dietary strategies. Posterior probability was calculated as a proportion of posterior samples where the rate difference exceeded zero.

Results

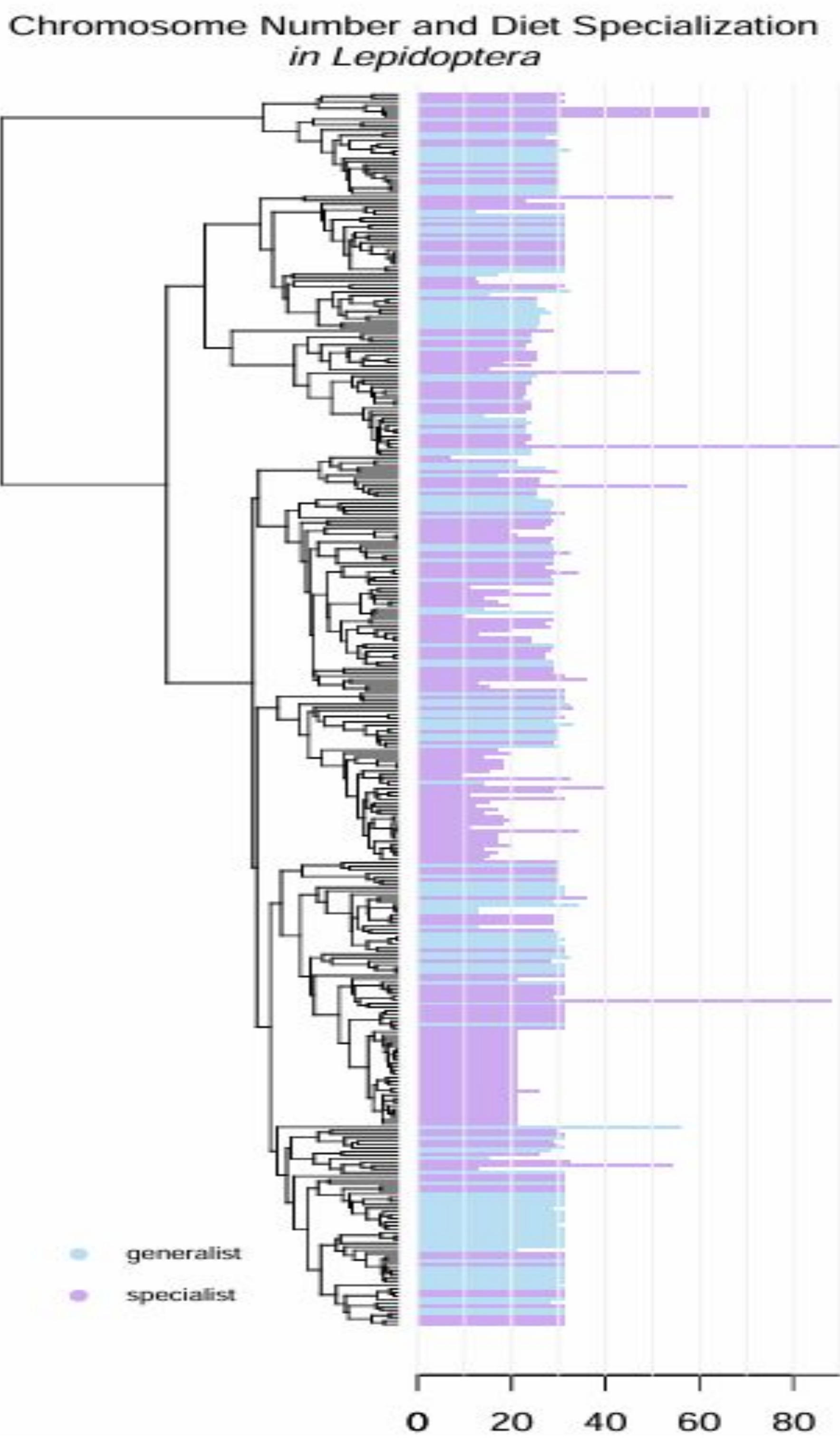
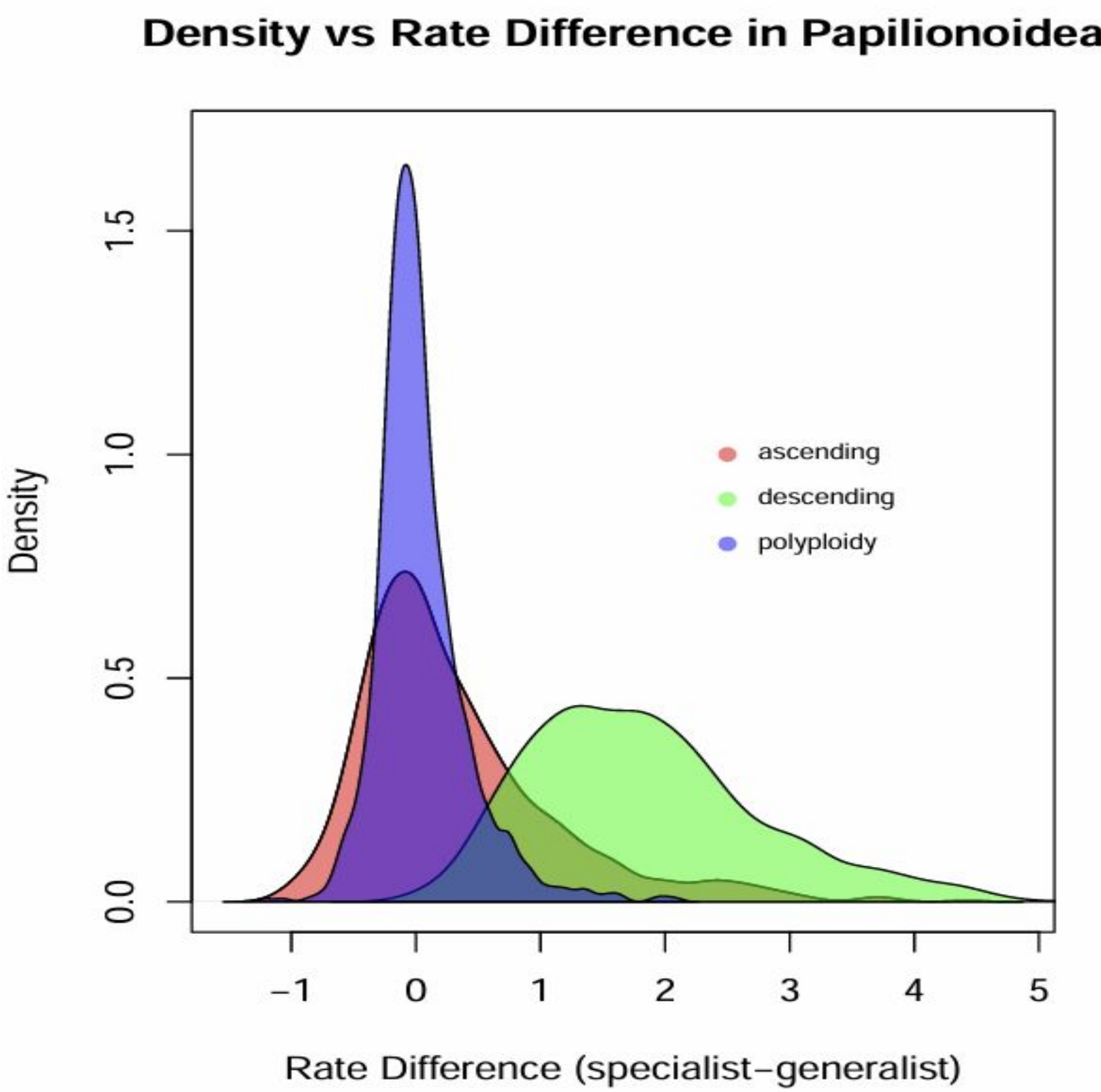


Figure 1. To the left is a phylogenetic tree of Papilionoidea showing distribution of haploid chromosomes and dietary strategies across species. The bar length represents haploid chromosome number (n) and the bar color indicates dietary strategy. Blue bars represent generalists and purple bars represent specialists. This visualization demonstrates the evolutionary distribution of both traits across the clade and provides the framework for comparative analysis.

Figure 2. To the right is a graph of posterior density distributions of rate differences in chromosome evolution between specialist and generalist Papilionoidea. Positive values indicate higher rates in specialists. Red denotes ascending changes (fissions), green denotes descending changes (fusions) and blue denotes polyploidy events. The dashed line indicates zero (no difference).



Parameter	Mean	Median	Lower 95 CI	Upper 95 CI	Prob Greater Than Zero
Ascending (Fission)	0.315	0.104	-0.726	2.509	0.587
Descending (Fusion)	1.842	1.729	0.439	4.037	0.999
Polyploidy	0.038	-0.024	-0.524	0.921	0.455

Table 1. The table above shows Bayesian MCMC results comparing rates of chromosomal change between specialist and generalist Papilionoidea. The parameters include ascending changes (fissions), descending changes (fusions), and polyploidy events. Rate differences were calculated by subtracting generalist rates from specialist rates for each parameter. The statistics reported include mean, median, 95% credible interval, and posterior probability that specialists exhibit higher rates than generalists.

Discussion

According to the results, dietary specialization significantly accelerates the rate of chromosome evolution in Papilionoidea. Fusions significantly increase the rate of chromosome evolution in specialists, while fissions and polyploidy rates remain the same between specialists and generalist. Based on the data from Table 1, fusions have a posterior probability value of 0.999 and a 95% credible interval of 0.439-4.037. On the other hand, fissions have a posterior probability of 0.587 and a 95% credible interval of -0.726-2.509. Polyploidy has a posterior probability of 0.455 and a 95% credible interval -0.524-0.921. The data suggests that specialists have a significantly higher rate of fusion than generalists, while there is no significant difference in the rate of fusion and polyploidy rates between specialists and generalists.

The data provides partial support for the hypothesis that diet specialization increases the rate of chromosome evolution. However, instead of a uniform increase across all types of chromosomal mutation, only fusions observed a significant change. This suggests that the relationship between ecology and genome evolution are more complex than previously expected. The elevated fusion rates in specialists support claims from the drift barrier hypothesis. Diet specialization is typically observed in smaller, isolated populations due to their dependence on disjunct patches of host plants. Genetic drift is has a considerable impact on these populations, especially compared to larger generalists populations because it can overpower negative selection against chromosomal rearrangements. This allows fusions that would normally be eliminated in larger populations to reach fixation. The chromosome distribution in figure 1 provides a foundation for detecting these different rates.

The fact that only fusions, and not fissions, indicates that elevated rates in specialists points to a potential adaptive mechanism. Chromosomal fusions link genes together, which prevents them from being separated in recombination. This may be beneficial in situation where a certain gene combination is advantageous. For example, in cases where specialists have to adapt to a chemically adapted host plant or for genes involved in detoxification, sequestration, and host recognition, a fusion of genes ensures that this gene combination is passed down together.

These findings are essential for understanding the drivers of biodiversity in Lepidoptera. The results suggest that dietary specialization can drive particular modes of chromosome evolution that facilitate reproductive isolation and speciation. Holocentric chromosomes and population dynamics created by diet specialization creates ideal conditions for rapid chromosome diversification.

Conclusions & Acknowledgements

- Dietary Specialization accelerates chromosome fusion rates, but not fusion or polyploidy rates.
- The results collected partially support the hypothesis and the drift barrier hypothesis.
- Chromosome fusions may provide an adaptive advantage for dietary specialists.

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